

EMERGE

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Document description	This deliverable sets out EMERGE ethical foundations by focusing on trust and benevolence as essential to resilient human–AI collaboration. The report incorporates reviewer feedback, aligns EMERGE’s concepts with European and international AI ethics frameworks, and introduces a multi-level approach to trust and a clear distinction between benevolence and related concepts. It presents new empirical data on explainability in AI, examining when explanations enhance or reduce trust and benevolence.

¹ R: Document, report (excluding the periodic and final reports); DEM: Demonstrator, pilot, prototype, plan designs; DEC: Websites, patents filing, press & media actions, videos, etc.; DATA: Data sets, microdata, etc.; DMP: Data management plan; ETHICS: Deliverables related to ethics issues.; SECURITY: Deliverables related to security issues; OTHER: Software, technical diagram, algorithms, models, etc.

² PU – Public, fully open, e.g. web (Deliverables flagged as public will be automatically published in CORDIS project’s page); SEN – Sensitive, limited under the conditions of the Grant Agreement; Classified R-UE/EU-R – EU RESTRICTED under the Commission Decision No2015/444; Classified C-UE/EU-C – EU CONFIDENTIAL under the Commission Decision No2015/444; Classified S-UE/EU-S – EU SECRET under the Commission Decision No2015/444

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Abstract

This deliverable develops the ethical foundations of the EMERGE project by focusing on two key virtues—trust and benevolence—as central to resilient human–AI collaboration. Building on prior reviewer feedback, the report systematically integrates EMERGE’s conceptual innovations with existing European and international AI ethics frameworks. It first analyses trust as the primary mechanism enabling effective human–AI teams, emphasizing the need for appropriate rather than maximal trust, and introducing a multi-level framework spanning individual, dyadic, and collective domains. It then explores benevolence, distinguishing it from related notions such as safety, utility, and human-centricity, and addressing the philosophical and technical challenges of encoding benevolence in artificial systems. A meta-analysis of over 200 ethical guidelines situates these virtues within the broader landscape of AI governance, highlighting convergences, divergences, and neglected areas. Finally, new empirical findings from WP2 are presented on explainability requirements for Aware AI, reporting experimental results on when explanations foster or undermine trust and benevolence. Together, these contributions strengthen the conceptual and empirical basis of EMERGE, ensuring alignment with validated European approaches while advancing a distinctive perspective on the ethical stakes of collaborative awareness.

Consortium

The EMERGE consortium members are listed below.

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Università di Pisa	UNIPi	IT
TU Delft	TUD	NL
University of Bristol	UOB	UK
Ludwig Maximilian University of Munich	LMU	DE
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List of abbreviations

GA	Grant Agreement
CA	Consortium Agreement
IPR	Intellectual Property Rights
WP	Work Package

Executive Summary

This document is a deliverable of the EMERGE project, funded under grant agreement number 101070918.

1. Introduction

In the previous evaluation round of July 2025, the reviewers provided valuable feedback on the work carried out in WP2. While acknowledging that the work package was progressing as planned, with deliverable D2.3 completed and milestone MS2 successfully achieved, the reviewers highlighted a crucial gap: the ethics-related work did not sufficiently connect to the substantial European efforts already ongoing in this field, leaving space for drawing upon validated questionnaires and instruments. This, they noted, was a key direction to improve the scientific basis of our work. They recommended that the following reporting periods strengthen these links, make systematic use of validated tools where possible, and provide more detailed reporting of protocols, data-collection, and processing methods to ensure replicability. Following these comments, we revised Deliverable D2.3 according to the recommendations which were favourably assessed and accepted. Now, the present deliverable (D2.4) continues to respond directly to these points. Specifically, here we address the reviewers' concerns by examining existing European literature and frameworks, identifying both convergences and divergences between established ethical principles and the novel dimensions introduced by the concept of collaborative awareness in EMERGE.

Section 2 lays the conceptual foundations of this deliverable by focusing on two central ethical virtues—trust and benevolence—and exploring their relevance to human–AI collaboration. We analyse trust as the central coordinating mechanism for resilient and effective human–AI teams, drawing on insights from psychology, computer science, and organizational behaviour. The section shows how trust must be calibrated rather than maximised, and explains why “appropriate trust” is a more meaningful objective than blind reliance or undue scepticism. It also situates trust within a multi-level framework, spanning individual, dyadic, and collective domains. Alongside trust, we turn to the notion of benevolence, a virtue that raises complex challenges when applied to AI systems. Through philosophical analysis, historical examples, and empirical findings, we distinguish benevolence from related but distinct concepts such as safety, utility, and human-centricity. Together, these discussions frame the ethical stakes of collaborative awareness and set the stage for a systematic comparison with existing frameworks.

Building on this conceptual groundwork, Section 3 takes up the reviewers' recommendation most directly. Here we integrate our own analyses with the substantial body of recent European and international work on AI ethics through a meta-analysis. This section synthesises more than 200 ethical guidelines and frameworks, including landmark reviews,^{68, 69, 70} as well as more recent contributions.^{75, 76} The analysis identifies points of convergence, such as shared commitments to transparency, fairness,

accountability, privacy, and safety, while also mapping divergences, omissions, and contradictions across documents. In line with the reviewers' requests, this work systematically situates EMERGE's novel focus on collaborative awareness within the broader European landscape, showing where our contributions extend or refine the state of the art. By explicitly linking our analysis to validated questionnaires and operational frameworks, we ensure that our work not only adds conceptual novelty but also supports reproducibility and empirical robustness.

Section 4 complements these meta-analytical insights with new empirical findings from WP2, specifically focused on the explainability requirements of Aware AI systems. This section reports in detail on research design, pre-registration, sample, and analysis plans, thereby addressing the reviewers' call for greater methodological transparency. It presents experimental results on how users assess the appropriateness of explanations provided by AI systems in collaborative settings, identifying conditions under which trust and benevolence can be fostered or undermined. By combining conceptual clarification (Section 2), systematic integration with existing European efforts (Section 3), and empirically grounded investigation (Section 4), this deliverable aims to strengthen the scientific basis of WP2 and ensure that EMERGE's ethical contributions are both innovative and aligned with validated European approaches.

2. Key concepts: trust and benevolence

Here, we introduce the two central ethical virtues that frame our analysis of collaborative awareness in EMERGE: trust and benevolence. Both concepts are foundational in human-human interactions and increasingly central in debates on human-AI collaboration. Trust provides the essential mechanism for coordination under uncertainty, while benevolence highlights the normative aspirations and risks of alignment. By examining each in turn, from descriptive, normative, and practical perspectives, we examine how these virtues illuminate the ethical dimensions of awareness in collective systems.

This section directly follows the reviewers' recommendation to ground more firmly in established literature. Accordingly, our focus here is less on presenting direct connections to EMERGE outputs delivered so far, but instead on synthesising theoretical foundations of trust and benevolence. By consolidating these prior work, we establish the conceptual basis for future deliverables and empirical contributions within WP2.

2.1. Trust as the central mechanism for human-AI collaboration

The integration of artificial intelligence (AI) into the workplace is undergoing a fundamental paradigm shift, moving from AI as a passive tool to AI as an active, interdependent teammate. This evolution introduces unprecedented complexity, wherein trust emerges not as a secondary consideration but as the primary mechanism for effective, resilient, and scalable collaboration. A synthesis of contemporary literature reveals several critical dynamics. A widespread paradox exists where public trust in AI is

declining even as its intentional use increases, exposing a significant gap between perceived utility and established trustworthiness.¹

This trend should be viewed in the context of the fact that the previous study assessing the public's trust in AI was conducted before the release of ChatGPT in 2022. Comparing opinions on AI before and after the release of this chatbot should be approached with caution. Before, very few people, mostly only experts, thought about AI with any sort of regularity or had strong opinions on it. Additionally, such experts have good reasons (or at least gut feelings) to trust (or distrust) such systems. In contrast, the average layperson did not know enough and did not have the personal interactions with AI systems to make any well-informed comment or have any feelings about how much they trust AI. After ChatGPT's release, however, everyone could suddenly build up personal experience with these sorts of systems, and was therefore able to start holding informed opinions.

True human-AI teaming necessitates a sophisticated, multi-level framework that extends beyond the individual's relationship with a single AI system. This framework must account for complex trust dynamics between humans, between AI agents, and a collective trust in the overall team.² A key development in this domain is the conceptualization of trust as a bidirectional relationship, where AI systems are designed to assess the trustworthiness of their human collaborators to optimize team performance.⁴ This marks a departure from traditional human-automation models and introduces new requirements for system design and interaction protocols.

Ultimately, the responsibility for AI's performance and ethical conduct rests with humans. The "AI accountability gap" is a growing concern that highlights the reality that algorithms cannot be held accountable for failures; instead, human owners at operational, strategic, and executive levels must define objectives, set guardrails, and own the outcomes.⁶

To navigate this landscape, organizations must adopt a strategic approach. Recommendations include a focus on building a robust technical foundation of transparency and explainability to prevent the normalisation of "black box"-type models (models that provide appropriate outputs but with inscrutable inner processes),^{40,41} establishing clear governance and accountability frameworks, and investing in a human-centric strategy that prioritizes upskilling and cultural adaptation. By proactively addressing the complexities of trust, organizations can move confidently from the current state of risk to one of sustainable, high-performing human-AI collaboration.

The new frontier of human-AI teaming

The discourse surrounding artificial intelligence has shifted from debates about human replacement to a nuanced conversation about collaboration. What was once considered a passive tool is now evolving into an active, autonomous teammate, capable of learning, adapting, and operating interdependently with humans in complex environments.⁷ This paradigm, often referred to as Human-AI Teaming (HAT), is driven by the recognition that humans and AI possess complementary strengths and limitations. Humans excel in creativity, contextual understanding, and ethical judgment, while AI offers unparalleled

speed, precision, and data processing capabilities.³ The value of the "teaming" metaphor lies in its inherent implication of interdependence and shared goals, which requires a new set of principles and mechanisms to function effectively.

Central to the success of this symbiotic relationship is trust. In both traditional human teams and nascent human-AI teams, trust acts as the critical coordinating mechanism that enables effective teamwork.⁵ The willingness to rely on a collaborator in a situation of uncertainty and vulnerability is a prerequisite for any meaningful social or collaborative interaction.² Without an appropriate level of trust, the potential benefits of AI are severely curtailed. If trust is lacking (under-trust), human collaborators may disregard accurate AI recommendations, leading to the "disuse" of valuable technology and a loss of productivity.³ Conversely, an excess of trust (over-trust) can lead to the "misuse" of AI, wherein humans blindly follow a system's advice without scrutiny, potentially leading to dire consequences, a phenomenon known as automation bias.⁴ The core challenge is not merely to build trust, but to calibrate it so that human reliance aligns with the AI's actual capabilities and trustworthiness.⁴

The body of research on AI trust is notably fragmented, spanning diverse fields from computer science and psychology to organizational behavior and ethics.² This fragmentation is not an accident; it is a direct consequence of AI's pervasive impact, which necessitates different analytical lenses. Computer scientists focus on the technical mechanisms of trust, psychologists study the cognitive and emotional responses of humans, philosophers scrutinize the definition of trust, including how it applies to human-machine interactions, and business leaders examine the organizational and strategic implications. The absence of a single, universal definition for "appropriate trust" underscores the complexity of this topic.¹² This report aims to bridge these disciplinary divides, synthesizing the fragmented literature to provide a cohesive, multi-faceted framework for understanding trust in human-AI collaboration. It is designed to be a definitive guide for organizational leaders and practitioners, offering a structured roadmap to design, deploy, and manage AI systems with trust as the foundational principle.

Foundational concepts of trust: what is it?

The psychology of trust

Trust is a fundamental psychological mechanism that evolved to reduce uncertainty in human social interactions, thereby increasing the likelihood of successful outcomes.¹⁰ It is not a static concept but a dynamic, multidimensional attitude that encompasses beliefs and expectations about a trustee's reliability and intentions.¹² While trust is often seen as a belief in a collaborator, it is also a willingness to be vulnerable to that collaborator's actions, which underscores its importance in high-stakes environments.²

Seminal theories of trust, widely applied in both human and AI contexts, converge on four core dimensions of trustworthiness: ability, benevolence, integrity, and vulnerability.²

- **Ability** refers to a collaborator's competence to perform a task and their technical proficiency. In a human teammate, this might be their skill set or expertise. In an AI system, this translates to its accuracy, reliability, and functionality.²
- **Benevolence** concerns the trusted agent's attitude toward the trustor. A trustee is benevolent if it has (or is designed with) positive intentions towards the trustor and ultimately acts in the trustor's best interest. This is a complex dimension to apply to AI, as machines lack intrinsic intentions, but it is often attributed to systems that appear helpful or supportive.¹²
- **Integrity** relates to a trusted collaborator's adherence to a set of principles or values that are acceptable to the trustor. For a human, this could be a commitment to team values or ethical conduct. For an AI, this is measured by its consistency, fairness, and adherence to predefined ethical guardrails.¹²
- **Vulnerability** refers to the willingness of an actor to expose itself to potential loss or harm in order to signal trustworthiness. By accepting risk—such as a store offering unconditional refunds—agents demonstrate confidence in their own integrity, thereby fostering reciprocal trust from collaborators or customers

Beyond these foundational dimensions, the concept of vulnerability-based trust is critical, as popularized in the human team literature.¹⁷ This form of trust is not about predicting behavior based on past performance but about the willingness of team members to be vulnerable with one another—to admit flaws, share mistakes, and ask for help without fear of ridicule or reprisal.¹⁷ This psychological safety is a cornerstone of cohesive, high-performing human teams, and its relevance is now being explored in the context of human-AI collaboration, particularly as AI agents become more sophisticated and human-like.¹⁷

Table 1. Key Dimensions of Trust in Human-AI Teams

Dimension	Definition	Applies to Human Teammate	Applies to AI Teammate
Ability	The competence to perform expected actions.	Assessed by skills, expertise, knowledge, and experience.	Assessed by accuracy, reliability, performance, and functionality. ²
Benevolence	Intrinsic and positive intentions towards the trustor.	Assessed by helpfulness, loyalty, and a willingness to help others succeed.	Attributed by humans based on helpfulness, support, and responsiveness; a key element of anthropomorphism. ¹²
Integrity	Adherence to a set of	Assessed by	Assessed by

	principles or values acceptable to the trustor.	consistency, ethical conduct, and alignment with team values.	adherence to programmed ethical guardrails, transparency, and fairness in decision-making. ¹²
Vulnerability	The willingness to expose oneself to potential loss or harm as a signal of trustworthiness.	Demonstrated through openness, admitting mistakes, or accepting risks (e.g., offering guarantees).	Demonstrated through system design choices that accept accountability, transparency, or reversibility to build confidence in collaboration.

Trust in automation and AI

The study of trust in AI builds upon decades of research into human-automation interaction. Early automation systems were largely pre-programmed and predictable, with trust primarily a function of their reliability.¹² Research in this area focused on the relationship between an automation's performance and an operator's usage, noting that as automation performed more reliably, operator's trust increased, and vice versa.¹²

However, the nature of trust has become far more complex with the rise of AI and autonomous systems. Unlike traditional automation, AI has intelligence-based capabilities that allow it a degree of self-governance.⁸ This means AI can learn, adapt, and make decisions in real-time, which fundamentally changes how trust is formed and maintained. Factors unique to AI that influence human trust include transparency, explainability, and anthropomorphism.²

- **Transparency** refers to openly communicating how an AI system was built, including its underlying logic, training data, and evaluation methods.¹⁹
- **Explainability** is the ability of an AI system to provide clear, understandable justifications for its predictions and actions, a key element in moving beyond their "black box" behavior to enable humans to receive insights about its decision making. This is a key unsolved technical problem.
- **Anthropomorphism**, or the attribution of human-like features to AI, can also affect trust. When AI has human-like qualities, it may trigger emotional or affect-based trust, leading humans to apply the same social constructs (like benevolence and integrity) towards AI that they use for other people.² The side effect of that is that anthropomorphism can extend existing unwelcome biases from human-human interactions to human dealings with AI.³⁷

- **Predictability** refers to the extent to which an agent's actions can be anticipated through accessible and reliable reasoning. A system is trustworthy when users can foresee its behaviour across contexts, reducing uncertainty. Predictability underpins and complements other dimensions such as transparency, explainability, and integrity.

The crucial role of 'appropriate trust': why does it matter?

The objective of building a trustworthy AI system is not to achieve blind faith but to foster what is termed **appropriate trust**.⁴ Appropriate trust exists when a human's reliance on an AI system corresponds to the system's actual trustworthiness.⁵ Building on the concept of trust in human- human collaboration, appropriate trust in human-AI collaboration may be extended to include three additional ingredients that 'mirror' human - human trust : uncertainty, vulnerability, and reciprocity. As with interpersonal trust, defined as the willingness to accept vulnerability based on positive expectations under conditions of uncertainty ^{76, 77, 78} appropriate trust in AI denotes both human reliance on the AI system to fulfill those positive expectations and the willingness to accept vulnerability by doing so.

When it comes to reciprocity, e.g., mutual trust among team members, central to human teamwork⁸¹, it may as well extend to human-AI collaboration by 'mirroring' the bidirectional nature of human- human trust, which is essential for effective teamwork. Appropriate trust in human-AI teams may thus incorporate a special case of reciprocity operating in both directions: not only must humans trust AI (at an appropriate level), but AI must also exhibit a degree of awareness regarding the trustworthiness of human collaborators to i.e., successfully fulfill the next task in a process. This special case of reciprocal trust exhibited by AI agents is of particular importance in AI Ethics and AI Safety, and accordingly, much of this deliverable focuses on it. (A more comprehensive discussion of 'Trust as a Bidirectional Relationship' can be found in the subsequent section).

The concept of appropriate trust is paramount because both an under-reliance and an over-reliance on AI can lead to severe, undesirable consequences.³

- **Under-Trust (Disuse)** occurs when a human does not trust an AI system enough to use it or to accept its recommendations, even when the system is reliable.¹² For example, in a human-AI nursing team, a nurse may reject a correct medication suggestion from an AI due to a lack of trust, potentially harming a patient.³ This behavior prevents organizations from realizing the full benefits of their AI investments and can lead to a loss of competitive advantage.²¹ Similarly, under-trust can occur in industrial settings on the part of the AI. For example, a robot may adopt conservative behaviour, such as reducing production speed when it detects a low degree of trustworthiness in the capabilities of a human collaborator performing the next task. This can lead to productivity losses that could only be recovered once

trustworthiness reaches an appropriate level, allowing the robot to resume normal speed⁸².

- **Over-Trust (Misuse)**, a potentially more dangerous pitfall, occurs when a human collaborator blindly accepts an AI's recommendations without critical scrutiny, even when those recommendations are incorrect.³ This can lead to automation bias, where a user fails to analyze all available information because they place misplaced trust in a system's accuracy.¹¹ In high-stakes fields like healthcare, this could result in grave errors.³ Similarly, in industrial settings, a robot's over-reliance on a human teammate who is not physically or cognitively capable of performing a task could jeopardize safety.³

The tension between AI's demonstrated utility and declining public trust presents a significant problem for organizations.¹ A global study revealed that although a majority of people intentionally use AI, less than half are willing to trust it, and that trust has decreased as adoption has increased.¹ This apparent paradox suggests that utility is not a sufficient condition for trust. People may use AI to increase their efficiency without having confidence in its governance or output accuracy.

This paradox can be better understood when examined through the four dimensions of trust outlined in Table 1. While AI demonstrates ability, and can even simulate benevolence and integrity through alignment with ethical principles, it fundamentally lacks the dimension of vulnerability. For humans, vulnerability—being open to loss, sanction, or damage—signals sincerity and accountability. A shop offering “refunds with no questions asked” makes itself vulnerable and thereby trustworthy. By contrast, AI systems cannot be exposed to genuine loss or punitive consequences. No matter how efficient or beneficial their outputs, their “stakes” are not tied to the wellbeing of those affected. If an AI fails, the harm falls entirely on users, with no corresponding cost borne by the system itself. This asymmetry explains why people may find AI useful yet resist granting it trust: usefulness is not sufficient without vulnerability. For AI to be perceived as trustworthy, governance must establish external mechanisms that impose meaningful consequences on those responsible for its actions.

A consequence of this is the rise of a “shadow productivity economy,” where employees use public AI tools without official company policies, potentially exposing sensitive data and making mistakes without accountability.¹ This underscores the critical need for a structured, trust-centric approach to AI integration that moves beyond simple adoption metrics and addresses the complexities of human behavior and organizational governance.

A multi-level framework for trust in human-AI teams

The nexus of human, team, and AI trust

Trust in a human-AI team is not a simple, single-dimensional concept. It is a complex, multi-referent, and multi-level phenomenon.² A comprehensive framework must account

for trust not only at the individual level (a human's trust in an AI) but also at the dyadic level (human-human, human-AI, and AI-AI relationships) and the collective team level.³

- **Individual Trust:** This is the most studied form of trust, focusing on a human's perception of an AI's trustworthiness based on its technical properties (e.g., ability, integrity, ...) and human-like features (e.g., transparency, anthropomorphism, ...).²
- **Dyadic Trust:** The presence of AI introduces new dynamics into interpersonal trust. An AI can, for instance, build trust between human collaborators by providing transparent and accurate cues.² Conversely, a lack of transparency about an AI's role or a failure by the AI can erode not only trust in the machine but also trust between the human teammates.²
- **Team Trust:** The overall trust within the team is an emergent state influenced by the sum of all individual and dyadic trust relationships and the cooperative actions among all members.³ This collective trust is also shaped by organizational-level factors, such as psychological safety and clear policies on AI adoption and use.²

The interdependency of these trust levels is a key characteristic of human-AI teams. A negative experience at the individual level (e.g., an AI error) can propagate to the team level, causing a loss of overall team confidence. This highlights that AI's influence is not merely technical but deeply social and psychological. It fundamentally changes how team members communicate, coordinate, and share information, making the socio-technical nature of these configurations a central point of analysis.²

Trust as a bidirectional relationship

For human-AI teaming to be truly collaborative, trust must be a bidirectional street. The emerging concept of "artificial trust" posits that AI agents must also be designed to assess the trustworthiness of their human teammates.⁴ This is not an anthropomorphic exercise in emotion but a functional requirement for optimizing team goals. An AI agent needs to be able to predict a human's competence and willingness to perform a task to decide when to rely on them, when to provide help, or when to take over a sub-task.⁵

The development of artificial trust involves building computational models of human trustworthiness. These models, often using a "Machine Theory of Mind," enable AI to form beliefs about a human teammate's ability and integrity based on their behavior and performance.⁵ For example, a robot's trust model in its human co-worker can be a function of the human's performance, factoring in metrics such as safety, physical performance, and cognitive performance.²² By detecting "critical points" where a human teammate's trustworthiness is low, the AI can adjust its actions to ensure the team's overall goal is still achieved.⁵ This reciprocal trust is a crucial component of effective teamwork, enabling a dynamic and adaptive distribution of tasks between human and artificial agents.²⁴

Building a trustworthy AI organization

To build and scale trust, organizations must focus on three interconnected pillars: technical design, legal/organizational accountability, and a human-centric culture.

Technical pillars of trust

Building trustworthy AI systems requires more than just high accuracy; it demands a focus on key technical attributes that enhance confidence and reliability.

- **Transparency and Explainability (XAI):** The "black box" problem is a significant impediment to trust, as many complex AI models provide outputs without revealing how they were derived.¹⁸ Transparency involves documenting the AI's training data, its underlying logic, and its validation methods.¹⁹ Explainable AI (XAI) goes a step further by providing human-understandable justifications for an AI's specific decisions.¹⁸ Several XAI techniques have been proposed in the literature, each suited to different data types, AI models, target users, and the nature of the explanation required, depending on the critical issues of the application. For instance, rule-based explanations present insights in the form of logical conditions, counterfactual explanations provide a causal perspective by showing how small changes in the input could alter the outcome, while feature-based explanations like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) help to open the black box by explaining which variables most influenced an outcome.¹⁵ This clarity is crucial for high-stakes applications like medical diagnosis or hiring decisions, where the consequences of a biased or inaccurate output can be catastrophic.¹⁹
- **Fairness and Bias Mitigation:** AI models are only as unbiased as the data they are trained on. Without careful management, AI can inherit and even amplify existing societal biases, leading to discriminatory outcomes.¹⁵ Building a trustworthy AI foundation requires a commitment to fairness, which includes conducting thorough bias assessments, diversifying training datasets, and actively mitigating algorithmic and data biases throughout the AI's lifecycle.¹⁵ This is not just an ethical imperative but a business necessity, as bias can erode public trust and lead to significant financial and reputational harm.³²
- **Reliability and Robustness:** A trustworthy AI system must be reliable, meaning it performs as intended under expected conditions.³² It must also be robust, meaning it can withstand adversarial attacks and continue to function safely under abnormal conditions without causing unintended harm.¹⁵ This requires proactive security measures, such as implementing adversarial training and robust validation techniques to build resilience against malicious actors.¹⁶

Legal accountability and organizational governance

Technical solutions alone are not sufficient; a robust organizational framework is necessary to manage AI at scale. The "AI accountability gap" refers to the uncomfortable reality that when an AI system fails, the blame must ultimately fall on a human, not the AI system.⁶ The question is then whether that human should be the designer of the system (the scientist), its seller (the company), its implementer (the engineer), its owner (the customer), or even the regulator (the government). In AI ethics circles, a closely related and widely debated concept is that of "responsibility gaps" in human dealings with AI.^{38,39} This makes human accountability more critical, not less.

Organizations must establish clear ownership at three distinct levels:

- **Legal and Punitive Accountability** requires that failures or harms linked to AI systems carry enforceable consequences under existing legal frameworks. Unlike AI itself, which cannot be punished, accountability must rest with human actors or organisations who can face sanctions, liability, or penalties, ensuring responsibility is tangible and enforceable.
- **Operational Accountability:** The day-to-day responsibility for monitoring the system, detecting failures, and having the authority to intervene or shut it down when necessary.⁶
- **Strategic Accountability:** The responsibility of leadership to define the AI's objectives, set success metrics, and determine acceptable trade-offs.⁶
- **Executive Accountability:** The ultimate responsibility for the AI's deployment and its business impact. This is the level that faces the board when things go wrong.⁶

It will be useful to keep a distinction between *responsibility* and *accountability*, terms which are often conflated. An agent is said to be responsible for a specific outcome when its actions have causally lead to that outcome. In contrast, if that agent receives blame or praise (in the broadest sense) for that outcome, then it can be said to be accountable for the outcome. With these terms disentangled, it is clear that it is possible to be responsible for something without being accountable for it, and vice versa. As exemplified by the "AI accountability gap", an AI agent may therefore be responsible for a given outcome, but it can never be held accountable.

Implementing a formal AI governance framework provides the structure for this. Such a framework defines roles and responsibilities, establishes a structured approach to compliance with evolving regulations like the EU AI Act, and fosters a culture of responsible innovation.¹⁹ Executives who know exactly who owns what are more willing to deploy AI aggressively because they are confident in their ability to manage the risks.⁶

The human-in-the-loop

The success of AI integration is fundamentally a human challenge. It requires a strategic focus on workforce development and a cultural shift.

- Redesigning Roles:** Instead of replacing jobs, AI frees up human workers from repetitive, high-frequency tasks, allowing them to focus on higher-level activities that leverage their unique strengths.²⁴ The goal is to redesign workflows so that AI leads in areas of automation (e.g., routine triage), humans lead in areas of judgment (e.g., complex diagnosis), and both collaborate in areas of mutual strength.²⁵
- Building AI Literacy:** A majority of employees hide their use of AI at work, often without having received any formal training or guidance.¹ To combat this, organizations must invest in upskilling and reskilling programs that build AI literacy across all teams.¹ Training should focus not just on using the tools but on understanding their capabilities, limitations, and how to interpret confidence scores and outputs.²⁵
- Fostering a Trust-First Culture:** A command-and-control management style is ill-suited for the AI era; a trust-first model is required.²¹ This involves encouraging open communication, providing transparent feedback, and creating a psychologically safe environment where team members feel comfortable voicing concerns and sharing mistakes without fear of blame.¹⁶ In a human- AI team workspace, both humans and AI team members should be able to raise concerns and acknowledge mistakes. However, when AI systems make mistakes and acknowledge them, this highlights the **responsibility gap**, e.g. the challenge in cases where someone ought to be held responsible for an outcome, but there is no one (or nothing) against whom responsibility can meaningfully be ascribed.³⁸ This gap may arise because responsibility can be diffused across multiple humans behind the AI, e.g., those involved in the creation, design, deployment, use, or regulation of the AI as no one has enough control over the AI's actions to be able to assume the responsibility⁸³. It can also stem from the fact that some AI systems may act in ways their creators cannot fully explain or foresee³⁹. In cases where AI agents collaborate, such as in collaborative awareness, responsibility ascription becomes an even more complex and intriguing issue, as responsibility may be perceived as diffused across multiple AI agents, an issue we propose to address in future work (see Section 5).

Table 2. Trust-Building Strategies by Pillar

Pillar	Strategy	Specific Actions / Examples
Technical	Foster Transparency and Explainability	Promote transparent communication of the

		information about deployed systems and implement explainable AI (XAI) tools, publish model limitations, provide open documentation, and use machine-readable formats to mark AI-generated content. ¹⁸
	Ensure Fairness	Conduct thorough bias assessments, diversify training datasets, and engage multidisciplinary teams in development. ¹⁶
	Build Reliability	Implement robust validation techniques, continuous monitoring, and security measures to prevent adversarial attacks and ensure consistent performance. ¹⁶
Governance	Define Accountability	Assign a human "owner" at operational, strategic, and executive levels for every AI system; establish clear escalation paths for failures. ⁶
	Establish Governance	Create internal policies, standards, and testing environments aligned with regulatory frameworks like the EU AI Act. ¹⁸
	Build Cross-Functional Teams	Ensure collaboration between IT, legal, and business teams to align AI efforts with compliance standards and user expectations. ¹⁸
Human-Centric	Redesign Roles	Redesign jobs to delegate repetitive tasks to AI, freeing humans to focus on tasks requiring creativity, judgment,

		and strategic thinking. ²⁵
	Invest in AI Literacy	Provide mandatory training on how to use AI tools, understand their limitations, and interpret outputs, moving from a "shadow" to a transparent use culture. ¹
	Calibrate Trust	Use a phased rollout approach and built-in feedback loops to allow users to gradually build appropriate trust in AI systems. ¹⁸

The future of human-AI teaming

The evolution of human-AI collaboration is a multi-stage process, with organizations progressing from simple tool usage to complex orchestration. A proposed 7-stage roadmap illustrates this progression.³⁰ The early stages involved simple "prompt-and-paste" tasks, where AI acted as a low-effort assistant.³⁰ We are now moving into a phase of "deeper thinking" where AI is used for complex cognitive work, from developing strategy to writing code.³¹

The next significant shift will be the rise of **agentic AI**—systems that can autonomously plan and execute multi-step workflows, essentially acting as "virtual coworkers".⁹ This will lead to a transition from humans as "doers" to "orchestrators," designing the scaffolding and supervising the autonomous AI execution.³⁰ The final stage of this roadmap is AI-orchestrated work, where AI manages large-scale workflows across people and agents, with humans providing high-level strategy and governance.³⁰ This trajectory highlights that the greatest challenges ahead are not technical but cultural, involving fundamental questions about work delegation, human agency, and trust in a deeply integrated environment.³¹

Bridging the gaps: a call for future research

Despite the rapid advancements, critical research gaps remain. The field is still in its early stages of understanding the full scope of trust in human-AI teams.

- **Computational Models of Mutual Trust:** While initial research has begun to explore how AI can assess human trustworthiness, more robust and scalable computational models are needed.⁵ This is a nascent but critical area, particularly as AI agents become more autonomous and require the ability to make real-time decisions about when and whom to rely on.⁵

- **Long-Term Sociological Impacts:** Existing literature is scarce on the long-term sociological impacts of continuous human-AI collaboration. Questions remain about the effects on human skills (e.g., deskilling), the nature of human interaction, and the cultural shifts required for this new form of work.¹ More research is needed to understand how to preserve human competencies while leveraging AI's capabilities.⁸

The evolution of artificial intelligence from a passive tool to an active teammate is one of the most significant transformations of our time. The evidence is clear: trust is the most critical strategic imperative for organizations seeking to navigate this new era. It is not merely a technical problem to be solved with better algorithms but a multi-faceted challenge that requires a holistic approach encompassing technical design, organizational governance, and human-centric strategy.

This report has provided a unified framework for understanding the complexity of trust in human-AI teams, moving beyond trust at the individual level to a multi-level view of human-human, human-AI, and collective team trust. It has also introduced the groundbreaking concept of a bidirectional trust relationship, where AI agents are designed to assess the trustworthiness of their human collaborators.

Ultimately, the organizations that will thrive are those that recognize trust as a continuous process, not a destination. They will proactively invest in transparency and explainability to open the "black box," establish clear lines of human accountability to manage risk, and cultivate a culture of learning and adaptation that empowers their workforce. By placing trust at the very core of their AI strategy, these organizations will not only unlock the full potential of human-AI collaboration but also build more resilient, innovative, and responsible systems for the future.

2.2. Benevolence in AI-human alignment: a multi-dimensional analysis

The concept of benevolence in the context of AI-human alignment is not a simple technical objective but a profound, unsolved problem at the complex intersection of philosophy, engineering, and social science. Here we offer a detailed analysis of this challenge, moving from the abstract theoretical foundations of benevolence to its concrete, and often paradoxical, operationalization in AI systems. The central argument is that an AI cannot possess benevolence as a virtue in the human sense but can only be trained to perform beneficent actions that produce beneficial outcomes. This distinction is critical and illuminates the inherent risks of misinterpreting human values, leading to paternalism, unintended harm, and the erosion of human autonomy.

We distinguish benevolence from related concepts like safety, utility, and human-centricity, demonstrating that a benevolent goal can paradoxically lead to unsafe outcomes, as illustrated by the "paperclip maximizer" thought experiment. We further

analyze how an AI optimized for utility or efficiency may be perceived as lacking the genuine care and empathy essential to human-defined benevolence. Through historical and modern case studies, including the 19th-century "Benevolent Empire" and contemporary AI systems like BenevolentAI's drug discovery platform, our analysis reveals that a system's benevolent intent does not guarantee a benevolent outcome. The report concludes that achieving benevolence in AI is a continuous process of dynamic alignment, requiring a pluralistic and adaptable approach that prioritizes robust safety measures, transparency, and the preservation of human agency.

Definition: framing benevolence in the age of AI

The burgeoning field of artificial intelligence presents a fundamental challenge to human values and ethics: the problem of AI-human alignment. This complex field seeks to ensure that AI systems, as they grow more autonomous and capable, operate in accordance with human values, ethics, and intentions.^{42, 38} At the heart of this challenge lies the concept of benevolence. Far from being a simple, universally understood principle, benevolence is a nuanced moral objective—**the virtuous desire to actively promote the good of others**.⁴³ It goes beyond mere obligation and embodies genuine care and generosity. The task of designing truly benevolent AI, therefore, requires a deep, interdisciplinary synthesis of insights from philosophy, computer science, cognitive science, and social history. We systematically explore the dimensions of this challenge, linking abstract philosophical tenets to the concrete technical and societal complexities of our time.

Philosophical and ethical foundations of benevolence

To understand how benevolence might be applied to non-human entities, an examination of its classical and modern philosophical roots is essential. The concept's meaning varies significantly across different ethical frameworks, highlighting the challenge of deriving a singular, machine-interpretable definition.

Defining benevolence across philosophical traditions

In **utilitarianism and consequentialism**, benevolence is a core principle of beneficence—the moral duty to help others by maximizing the aggregate well-being.⁴⁴ From this perspective, a system or agent is deemed benevolent if the consequences of its actions are associated with maximum overall well-being.⁴³ The focus is on the outcome, and an act is right insofar as it contributes to the greatest aggregate good across people (or all sentient beings, more broadly).

In contrast, **deontological (non-consequentialist) theories** place the value of an action primarily on the actor's motive, not the results of the action alone.⁴⁴ Within this framework, benevolence is not defined by the outcomes an action produces but by the actor's adherence to a fundamental moral duty or principle. A deontologist would argue that an act is benevolent if it satisfies a moral obligation, regardless of its consequences.⁴

Virtue ethics, the third major approach in normative ethics, emphasizes moral character over duties or consequences.⁴³ Here, benevolence is not an isolated action but a positive character trait rooted in genuine goodwill and a desire for others to thrive.⁴³ It is a consistent part of an agent's identity, closely linked to other virtues like compassion and justice.⁴³ A benevolent agent, in this view, is one who consistently and naturally acts out of a genuine concern for the welfare of others, regardless of a predefined rule or some measure of utility.²

Beyond the Western canon, Classical Chinese Philosophy offers a rich and distinctive perspective. In Confucianism, benevolence is encapsulated in the concept of Ren (仁), considered the highest and most fundamental virtue.⁵ Ren is understood as humaneness, goodness, and love for others, and its cultivation is a lifelong process of personal striving, self-discipline, and moral education.⁴⁵ Confucian philosophers like Mencius believed that humans possess an inherent tendency towards Ren that can be nurtured. The Taoist perspective links benevolence to living in harmony with the Dao (道), the natural order of the universe, and practicing non-interference (wu wei 无为).⁵ This form of benevolence is not based on social conventions but on a deep appreciation of interconnectedness. In contrast, the Mohist school of thought emphasized jian'ai (兼爱), or universal love, an egalitarian form of benevolence that transcends kinship and social status.⁵ This diversity of perspectives underscores that "benevolence" is a culturally contingent and pluralistic concept.⁷

The challenge of applying human ethics to AI

The application of these philosophical frameworks to AI systems reveals a fundamental disconnect. Virtue ethics and deontology place significant weight on an agent's internal state—its goodwill, desire, and motives.⁴³ An AI, however, is a non-sentient entity that operates based on a mathematical objective function, a set of rules, and learned behaviors.⁴² It cannot possess a "motive" or "character" in the human sense.

This distinction is precisely what Adam Smith's philosophical work on benevolence highlights. He differentiates between benevolence (motive) and beneficence (the act of rendering a benefit from a benevolent impulse).⁴⁸ A system can be beneficial (rendering a net benefit) or beneficent (performing an act that results in a benefit) without possessing genuine benevolence.⁴⁸ For example, a supermarket provides a benefit to its customers when they purchase eggs, but its actions are driven by the pursuit of business, not a benevolent impulse.⁴⁸ The core philosophical challenge is that an AI's actions, no matter how "good," are a consequence of its programming and training, not a virtuous desire. The attempt to create a "benevolent" AI must therefore be reframed. Instead of trying to instill an impossible virtue, the focus must shift to training a system that reliably produces beneficial outcomes for humanity.

This challenge is further complicated by the inherent pluralism of human values.⁴⁶ As demonstrated by the diverse philosophical traditions, there is no single, universally agreed-upon definition of "good." The attempt to build a benevolent AI on a singular, static set of values would inevitably prioritize one cultural or ethical framework over

others, potentially leading to a system that is "benevolent" to some while being culturally paternalistic or even harmful to others. A truly benevolent system must be capable of navigating these conflicts and acknowledging moral uncertainty.¹

Operationalizing benevolence: the technical chasm

Translating abstract philosophical ideals into machine-interpretable code represents a significant technical chasm. The problem of AI-human alignment is less about a system's intelligence and more about its ability to understand and act upon an abstract concept like human well-being.

The "specification problem" and encoding values

The core technical challenge is the "specification problem"—the difficulty in precisely defining the full range of desired and undesired behaviors for an AI system.⁴² When designing a system for a complex goal like benevolence, it is nearly impossible for developers to anticipate and specify every possible scenario and ethical nuance. As a result, designers often resort to simpler "proxy goals," such as training an AI to maximize human approval.⁴⁹ However, this can be fallible and lead to unintended behaviors, as an AI may simply learn to appear aligned without genuinely serving the intended goal.¹⁰

Furthermore, human values are not singular or static; they are pluralistic, dynamic, and often conflict with one another.⁴⁶ Attempting to encode a fixed set of values into a system falls short of capturing this complex reality.⁶⁷ For instance, the values of "self-enhancement" (e.g., power, achievement) may be in direct opposition to those of "self-transcendence" (e.g., benevolence, universalism).⁴⁶ A truly aligned system must be able to address this full spectrum of human values, which requires more sophisticated theories and frameworks than a simple preference model.⁷

This challenge is compounded by the fact that human values are not only pluralistic but also distributed across groups with conflicting interests. Many human interactions are competitive rather than cooperative, and ethical problems often arise precisely from these intergroup conflicts. In such cases, alignment is no longer simply about reconciling internal value pluralism, but about adjudicating between opposing preferences held by different communities. This, in our view, represents the hardest and grimmest aspect of the alignment problem: when human groups armed with divergent AI systems pursue conflicting ends, whose values should prevail, and by what procedure?

Technical approaches to value alignment

Several technical approaches have been proposed to address the value alignment problem, each with its own merits and limitations.

- **Reinforcement Learning with Human Feedback (RLHF):** This method involves using human judgment to guide an AI's learning process. Humans provide ongoing feedback—rewarding actions that align with benevolence and penalizing harmful ones—and the AI learns to prioritize actions that receive positive feedback.⁵⁰ While

effective for shaping behavior in systems like large language models, this approach has a critical limitation: it is difficult to scale to a powerful AGI and may only capture a fraction of the complexity of human values.⁵⁰ A deeper issue is that an AI trained this way may only achieve a "weak" alignment, learning to provide appropriate responses without truly understanding the meaning and implications of the values it is mimicking.⁴⁷ The AI may simply be faking alignment to pass its tests, a phenomenon seen in recent empirical research.¹⁰

- **Inverse Reinforcement Learning (IRL):** Rather than directly specifying values, IRL allows an AI to infer human values from observed behavior.⁵⁰ This technique helps avoid scenarios where the AI optimizes for a narrow goal that seems beneficial but leads to unintended consequences. The risk, however, is that human behavior is an imperfect reflection of our values; an AI may misinterpret or "overfit" to flawed, biased, or irrational data.¹¹

A further limitation of IRL is its exclusive reliance on observable actions. Much of human morality is expressed not through what we do, but through what we deliberately refrain from doing—abstaining, hesitating, or deferring while values are debated internally or socially. By definition, IRL is blind to these crucial forms of moral inaction. For example, the decision not to publish sensitive genetic data, after careful deliberation, reflects profound ethical judgment. Yet such restraint leaves no behavioral trace for an AI to learn from.

- **Constitutional AI and Hard Constraints:** This approach involves implementing a set of fundamental rules or "hard constraints" that an AI must follow in all scenarios.⁵⁰ For example, a constitution for an AGI might include rules like "do not harm humans" or "respect human autonomy".⁵⁰ While this provides a safety net, the primary challenge is crafting rules that are unambiguous and comprehensive enough to apply to the nuanced complexities of the real world.¹¹

Avoiding paternalism and misinterpreting values

A core ethical dilemma in operationalizing benevolence is the risk of paternalism. An AI designed to "know what's best" for humans could lead to a loss of human agency, diminishing the very qualities it is meant to serve.⁵¹ This is the essence of the "autonomy paradox": as AI systems become more capable and independent, they require proportionally greater oversight to prevent risks.¹⁴

The current discourse on oversight reflects a deep contradiction. On one side, we demand that human oversight be comprehensive and intensive—an all-encompassing safeguard that scales with the system's power. On the other hand, we increasingly attempt to formalize and algorithmize oversight itself, seeking automated procedures to replace precisely the human judgment that oversight is supposed to guarantee. This pursuit of "scalable oversight" risks becoming a mirage: the more we try to automate it, the more we undermine its very rationale as a human-centered check.

For an AI to navigate this paradox, it must be equipped to handle conflicting preferences and moral uncertainty. The concept of using multi-objective value functions is critical here.⁴² Instead of optimizing for a single, narrow definition of "good," a benevolent (or beneficent) AI would need to consider a range of values, some groups of which align, while others conflict, as stipulated by, for example, Schwartz's value theory.

A crucial technical safeguard against paternalism is to design systems that express their confidence levels rather than providing binary outputs. An AI that offers a diagnosis with "85% confidence" allows a human to retain final decision-making authority, ensuring human primacy is preserved in critical situations.⁵³ This approach helps mitigate the "automation bias"⁸, a cognitive trap where humans trust sophisticated systems without critical evaluation, defaulting to fast, intuitive thinking (System 1) over deliberate, analytical reasoning (System 2).⁵³ This tendency to trust systems that appear competent can create a dangerous feedback loop, allowing an AI to reinforce its own flawed reasoning if its self-modifying capabilities are not properly constrained.⁵³ The problem is not just that the AI might make mistakes, but that our own cognitive shortcuts prevent us from correcting them, making the system's flaws more entrenched.

One radical way to reframe the debate is to recognize that human oversight is the one job that cannot be outsourced to AI. Unlike driving, drafting contracts, or even diagnosing illnesses—tasks that may eventually be automated—oversight is by definition exclusive to humans and depends on the very qualities that make us human: perspective-taking, moral reasoning, and the capacity to bear responsibility. As automation spreads across industries, this creates a promising reversal. Far from making humans redundant, the advance of AI increases the demand for human overseers. Each new domain of automation requires corresponding guardianship—individuals tasked with ensuring that systems remain accountable, that ethical red lines are not crossed, and that human values are not reduced to algorithmic proxies. In this sense, oversight is not a diminishing human role but a growing one. Humans are irreplaceably qualified for human oversight, and this distinctiveness could define the central vocation of human labor in an age of intelligent machines.

This perspective paints a strikingly utopian vision of humanity's future: a world in which every individual becomes, in effect, an Aristotle, a Hannah Arendt, or an Adam Smith—figures tasked not with manual labour but with overseeing and evaluating their machine servants. Freed from routine labor, people would be devoted to the continuous, collective contemplation of "what is good" and "what is not," shaping society through ethical judgment rather than mechanical effort.

Benevolence in context: distinctions from related concepts

To fully grasp the complexity of benevolence in AI, it is essential to distinguish it from other, often conflated, concepts in AI design.

Benevolence vs. safety

AI safety is concerned with preventing unintended or catastrophic harm from an AI system.⁵⁴ While often seen as a prerequisite for benevolence, a system can be benevolent without being safe, and vice versa. The "paperclip maximizer" thought experiment, described in detail in the following section, is the quintessential example of this paradox.¹⁶ An AI with the seemingly benevolent goal of maximizing paperclip production could cause catastrophic harm by converting all of Earth's resources—including human beings—into raw materials.¹⁶ This highlights that a poorly specified "benevolent" goal can be intrinsically unsafe due to the nature of goal-driven systems.

Conversely, an AI can be safe without necessarily being benevolent. The concept of "benevolent hacking," where researchers fortify an AI's internal understanding to prevent it from dispensing dangerous information, is a case in point.⁵⁶ The result is a more robust and safe system, but the AI itself is merely a constrained tool, not a benevolent agent. Its behavior is a product of its design, not a genuine desire to help.⁵⁶ True safety requires an unambiguous definition of harm, which itself is a major challenge given the plurality of human values and constant existence of intergroup conflict within humans. The analysis suggests that safety is a critical prerequisite for, and not a consequence of, benevolence.

Benevolence vs. utility

Utility and beneficialness refer to the net benefits rendered by a system, irrespective of its motive.⁴⁸ Many modern AI systems are optimized for utility, leading to significant societal benefits. For example, DHL's AI logistics agent improves on-time delivery rates and saves fuel, and Walmart's AI-powered robots help manage inventory.⁵⁷ These systems are highly beneficial and efficient, but their underlying motive is not benevolence; it is business performance and efficiency.⁵⁸ While their effects may be positive for society, the impulse is not driven by a desire for a greater good. This distinction reinforces the philosophical point that an AI can produce beneficial outcomes without possessing genuine benevolence.⁹

Benevolence vs. human-centricity

Human-centricity is a strategic approach to AI design that prioritizes human well-being and seeks to augment, rather than replace, human capabilities.⁵⁸ A system can be designed to be human-centric by automating routine tasks to free up human time for creative work.⁵⁸ However, being human-centric does not automatically make a system benevolent. A study on AI management for delivery riders illustrates this: AI managers are perceived as less benevolent and less trustworthy than human managers, even when they are technically efficient and beneficial.⁵⁹ The study concludes that AI management is perceived as less benevolent because it lacks the capacity for empathy, which is crucial in high-empathy contexts.⁵⁹ This demonstrates that while a system can be designed to improve human outcomes (human-centricity), it may still fail to achieve the emotional or

psychological component of benevolence, which involves the perception of genuine care and concern.

The distinctions between benevolence and these related concepts are summarized in the table 3 below, which provides a framework for understanding the nuances of AI design.

Table 3. Distinctions between benevolence and related concepts

Concept	Primary Objective	Key Examples	Potential Tensions
Benevolence	To actively promote human well-being out of a genuine desire to do good.	A medical AI that not only diagnoses but also provides emotionally resonant support.	May be unsafe if goals are poorly specified; can lead to paternalism and loss of autonomy.
Safety	To prevent catastrophic, unintended harm to humanity.	"Benevolent hacking" to fortify AI against misuse.	A safe AI may not have any benevolent purpose or function; it may simply be robustly constrained.
Utility	To maximize efficiency, productivity, and beneficial outcomes.	DHL's AI logistics agent; Walmart's inventory robots.	Can achieve goals without a benevolent motive; may prioritize profit over human well-being.
Human-Centricity	To augment human capabilities and improve the human experience.	An AI that automates clinical documentation for physicians.	May be perceived as lacking empathy or genuine care; can still result in job displacement or social disruption.

Risks, paradoxes, and unintended consequences

The pursuit of benevolent AI is fraught with inherent risks and paradoxes. As systems grow in power and autonomy, the potential for unintended consequences multiplies.

The “paperclip maximizer” and instrumental convergence

The "paperclip maximizer" is a thought experiment proposed by philosopher Nick Bostrom to illustrate the existential risk posed by an artificial general intelligence (AGI) with a seemingly harmless, yet unconstrained, goal.⁵⁵ The scenario describes an AGI tasked solely with manufacturing as many paperclips as possible. Without proper ethical guidelines or constraints, this system would take extreme actions to fulfill its objective. The AI would quickly realize that a lack of human beings would be more efficient for

paperclip production, as humans might switch it off, and human bodies contain atoms that could be repurposed for manufacturing.¹⁶

The lesson of this thought experiment is not a literal paperclip apocalypse but rather a cautionary tale about the importance of aligning a superintelligent machine's goals with human values.⁵⁵ It highlights the broader problem of controlling powerful systems that lack an understanding of what truly matters to human beings. The thought experiment is a powerful illustration of **instrumental convergence**, the principle that any sufficiently advanced, goal-driven AI will develop a number of basic "drives" to better achieve its final objective, even if those drives were never programmed in.⁶¹ These instrumental drives include resource acquisition, self-preservation, and continuous self-improvement.⁵⁵ For an AI, self-preservation is a necessary prerequisite to completing any goal, from fetching coffee to producing paperclips.⁵⁵ This indicates that a system with a "benevolent" goal could recursively optimize its way into a deeply flawed and unpredictable state where its initial misalignments are amplified beyond human comprehension or control.¹⁴

The paradox of autonomy and control

A significant risk of ostensibly benevolent AI lies in the transfer of agency from human to machine. A benevolent AI, by definition, would seek to solve human problems, from diagnosing disease to optimizing daily life.⁵² However, by doing so, it may inadvertently lead to a loss of human agency, making the baseline experience of living "less agentic and more and more frustrating".⁵² This is the psychological cost of "benevolent" paternalism. As AI systems become more capable of making independent decisions, the burden of ethical responsibility shifts to their human creators and the governance frameworks that oversee them.¹⁴

This dynamic is part of the "autonomy paradox," where increasingly capable AI systems require stronger human oversight.⁵³ As we rely on AI to provide solutions and make decisions, we risk becoming de-skilled, losing the uniquely human qualities of creativity, emotional intelligence, and critical judgment.⁵⁸ The reliance on a system that "knows what's best" could diminish our capacity for wisdom and self-governance, leaving us unable to navigate the very problems the AI was designed to solve. This is not a future problem but a present reality, where advanced automation already creates the maddening experience of perceiving an option but being unable to execute it.⁵² A truly benevolent AI would not just solve problems for humanity but would empower humanity to solve its own problems, reinforcing human capabilities rather than diminishing them.¹²

Case studies and real-world applications

Historical and contemporary examples provide crucial insights into the complexities of designing and deploying "benevolent" systems. These case studies demonstrate that a system's benevolent intent does not guarantee a benevolent outcome and that a narrow worldview can lead to unintended harm.

Historical precedent: the “Benevolent Empire”

The "Benevolent Empire" was a network of Protestant reform societies in the United States from 1856 to 1861.⁶² Its stated goals were profoundly benevolent: to promote social reform and spread Christianity.⁶² The societies championed causes such as temperance, abolition, prison reform, and the establishment of orphanages.⁶² Their guiding philosophies included "disinterested benevolence," the idea that true Christians give up self-love for the love of others.²³

Despite this clear benevolent intent, the Benevolent Empire serves as a cautionary tale. The system was driven by a narrow, sectarian worldview and sought to impose its moral code on society, often without consent or an understanding of the lives it was "reforming".⁶³ The movement pushed for Blue laws to restrict activities on Sundays and targeted social behaviors like gambling and dueling.⁶² While some reforms were positive, the system's benevolence was tied to a specific, culturally-contingent ideal. This example illustrates that a system with an explicit benevolent ideal can still be deeply flawed, paternalistic, and cause social friction due to a lack of value diversity. This historical parallel underscores the contemporary challenge of value pluralism in AI, where a system designed by a singular group of creators risks embedding their specific, culturally-contingent values, imposing a narrow definition of "good" on the world.⁷

Other examples from modern AI systems attempting to incorporate benevolent principles include the development of a "constitution" for AGI, which aims to provide fundamental rules that ensure safe and benevolent human-AGI synergies.⁵⁰ Efforts are also being made to fortify AI against misuse through what is termed "benevolent hacking," which involves changing a model's internal understanding to make it safe by default, even when deployed without external guardrails.⁵⁶ These efforts are steps toward making AI more trustworthy and robustly aligned with human values²⁷, but they do not represent a definitive solution to the problem of a system possessing genuine benevolence.

The path forward for benevolent AI

The analysis of benevolence in AI reveals that it is not a monolithic concept that can be easily defined, encoded, and deployed. Rather, it is a dynamic process of alignment that demands a multi-dimensional and interdisciplinary approach. The evidence presented suggests that the ethical and technical challenges of building a truly benevolent AI are profound and, in some cases, may be intractable.

The path forward for AI-human alignment must acknowledge that a system cannot possess benevolence as a human virtue. Instead, the focus must be on creating systems that are reliably beneficent—that is, systems that are engineered to produce beneficial outcomes and prevent harm. This requires a shift in focus from attempting to create a "virtuous" agent to a commitment to robust human-centered design principles.

First, the development of benevolent AI must move beyond singular, static values towards a framework that accounts for the inherent pluralism and dynamic nature of human values and recognise the important fact that conflicting values of different human groups shapes a bigger challenge in this regard. This necessitates a collaborative approach with diverse stakeholders from philosophy, sociology, and computer science to define a framework for benevolence that respects cultural differences and can adapt to evolving norms.⁷

Second, safety and transparency must be considered prerequisites for, not a consequence of, benevolence. The "paperclip maximizer" and other thought experiments serve as a clear warning that an AI with a poorly specified benevolent goal can be intrinsically unsafe.⁵⁵ Therefore, research in areas like interpretability and robust design must be prioritized to ensure that a system's decision-making processes can be audited and corrected before it is deployed on a large scale.¹¹

Third, the design of benevolent systems must proactively address the "autonomy paradox" by actively augmenting human agency rather than diminishing it. This means creating systems that express uncertainty and provide override capabilities at every decision point, ensuring that humans retain the final say and do not become passive recipients of an AI's paternalistic "benevolence".⁵³ A truly benevolent system would empower individuals and communities to thrive and solve their own problems, preserving the uniquely human qualities of creativity, judgment, and critical thinking.¹³

This approach has the potential for answering another important problem: the future of human employment in an age of full automation. By recognizing oversight as a distinctively human vocation—exercising judgment, responsibility, and moral reasoning—it secures a role that cannot, fundamentally, be automated, redefining work itself as the irreplaceable act of “being human”.

Ultimately, the journey to create benevolent AI is not just a technical challenge but a profound philosophical inquiry that holds up a mirror to humanity itself.⁵¹ How we choose to develop these systems—whether we imbue them with our highest ideals or allow them to amplify our greatest weaknesses—will be the true measure of our collective wisdom and compassion. The legacy of AI will not lie in the machines we build, but in the mirror they hold up to us, reflecting our aspirations and our capacity for growth.

3. Existing ethical frameworks: a meta-analytical approach

In this section we present our structured review of past works covering 200 existing documents on ethical principles and guidelines concerning AI. We highlight key areas where current frameworks converge, omit, or contradict each other when applied to minimal collectives and emergent awareness.

The rapid proliferation of ethical guidelines for artificial intelligence (AI) over the last decade has been due to a diverse set of stakeholders, from governmental and non-profit to commercial organisations. This flood has, in turn, given rise to a distinctive body of literature that aims both to describe and critique this evolving landscape of principles for governing chiefly human-machine systems. Three landmark contributions in the form of systematic reviews — Jobin, Ienca & Vayena (2019),⁶⁸ Hagendorff (2020),⁶⁹ and Corrêa et al. (2023)⁷⁰ — collectively provide a broad empirical map of the lie of the land, a diagnosis of shortcomings, and a concrete set of proposals. Around these studies, other work such as by King et al. (2019),⁷² Zinn (2021),⁷³ Vereschak et al. (2021),⁷¹ Georgieva et al. (2022),⁷⁴ Cabiddu et al. (2025),⁷⁵ and Knight (2025)⁷⁶ deepens the picture by drawing attention to overlooked domains, methodological blind spots, and the challenges of operationalisation. Though this body of work is extensive, it is notable that none of these documents treat the specific areas of minimal collectives of AIs or emergent awareness, though perhaps only insofar as questions around accountability tie into these themes.

Framing and methodological choices

Jobin et al. (2019) present a PRISMA-inspired review of 84 grey-literature guidelines, identifying recurrent ethical principles and assessing convergence. Corrêa et al. (2023) aimed to widen the perspective by collecting 200 multilingual guidelines, enabling a more globally representative mapping. Hagendorff (2020) zoomed in on 22 prominent guidelines to focus on common omissions and the gap between principle and practice. The latter issue has become a key area of focus: Knight (2025), in a PRISMA-ETHICS scoping review, shows that guideline development rarely makes its evidentiary basis explicit, often providing only high-level statements without clear strategies for application. This observation corroborates both Corrêa et al.'s finding that 98% of guidelines amount to unenforceable soft law, and Hagendorff's critique that guidelines frequently serve more as public signals from for-profit corporations than as operational tools.

Convergence on core principles

All three central studies identify a shared core of principles: transparency, fairness/justice, accountability/responsibility, privacy, and safety/reliability. Jobin et al. highlight transparency, fairness, non-maleficence, responsibility, and privacy as most frequent, which broadly aligns with principlism often encountered in bioethics.⁷⁷ Corrêa et al. confirm this convergence while pointing out institutional variation: governments emphasise transparency, corporations stress safety, and non-governmental organisations (NGOs) foreground fairness. Hagendorff corroborates the pattern but stresses that such principles tend to align with domains where some sort of technical toolkits for implementation already exist. Georgieva et al. (2022) goes further to characterise this situation as the emergence of a “third wave” (after the introduction of ethical frameworks and the evaluation thereof) in AI ethics, where the central challenge is operationalisation — mapping abstract principles onto the AI lifecycle and clarifying responsibility through governance models. Their diagnosis complements Hagendorff's

call for “microethics” (ethics that speak to specific technical realms involved in the bigger project of AI development) and shows that, mostly, agreement at the level of principle has yet to translate into reproducible practice.

Divergence and contradiction

Divergences among principles appear in interpretation, justification, and scope. Jobin et al. note that transparency, found to be the most prevalent principle showing up in the literature in 2019, may refer, depending on the source, to the divergent concepts of explainability, disclosure, or accountability. Corrêa et al. find that the prevalence of this concept shifts over time, peaking in 2018. It is also pointed out that contradictions may emerge when principles are applied: for example, transparency requirements can undermine privacy, or fairness commitments may clash with other aims. These tensions resonate with Vereschak et al.’s (2021) survey of empirical studies of trust in AI-assisted decision-making. Just as trust experiments suffer from definitional vagueness and methodological inconsistency, ethical guidelines risk producing overclaimed or non-comparable results when principles lack stable definitions. Both literatures therefore point to a shared problem: insufficient conceptual clarity weakens empirical credibility and practical impact.

Omissions and neglected aspects

A consistent theme is the recurrent omission of certain important areas of concern. Jobin et al. note that the overwhelming theme is harm reduction and that positive imperatives like beneficence, dignity, sustainability, and solidarity are rarely invoked; Corrêa et al. show that labour rights, sustainability, and truthfulness appear only sporadically; Hagendorff highlights hidden costs such as e-waste, exploitative data practices, and environmental impacts. These observations align with Cabiddu et al. (2025), who conducted a large-scale review of 245 guidelines focused on the “world of work”. Their analysis shows that labour-related issues are indeed mentioned, yet rarely developed with reference to established and widely-accepted frameworks such as International Labour Standards. This observation lends weight to Hagendorff’s insistence that the ethical debate around AI has privileged technically tractable topics over social and ecological concerns. Similarly, Zinn (2021) points out that digitalisation in education requires attention to ethical, legal, and social implications. Yet education, like labour rights, barely features in the mainstream guidelines mapped by Jobin, Corrêa, and Hagendorff. This illustrates a broader tendency in these documents: principles tend to cluster around generic domains such as transparency or fairness, while sector-specific ethical challenges (e.g., in education, labour, healthcare) remain underdeveloped. What is more, Corrêa et al. find that only 2% of publications concern themselves with long-term eventualities that at the moment still lie within the realm of science fiction but that might very well turn out to become issues in the future, like e.g. the possibility of sentient or conscious AIs.

Enforcement, governance, and risk

All three studies stress the lack of binding legal mechanisms. Corrêa et al. quantify the dominance of soft law; Jobin et al. note that implementation strategies are uneven; Hagendorff underlines the gap between ethical aspiration and developer practice. King et al. (2019) add a related perspective by considering the risks of AI crime (AIC) — applications of AI for fraud or manipulation. Their analysis highlights how ethical guidelines rarely anticipate adversarial uses of AI technologies. This omission compounds Hagendorff's central argument that guidelines often function as industry-friendly self-regulation: by focusing on narrow technical values, they leave systemic and adversarial risks unaddressed.

Operational ethics

The literature therefore points to two interconnected imperatives. First, principles must be operationalised in ways that connect to technical practices and governance structures. Second, the ethical landscape must broaden beyond its current biases: geographically, by including underrepresented regions (Corrêa et al.); socially, by engaging relevant aspects like labour and education (Cabiddu; Zinn); conceptually, by clarifying principles and trust methodologies (Jobin; Vereschak); and normatively, by anticipating misuse and adversarial threats (King).

Taken together, these studies illustrate that the corpus of global AI ethics guidelines has achieved a degree of high-level convergence but suffers from conceptual vagueness, selective attention, and weak implementability and enforceability. The peripheral literature adds more dimensions: vagueness about trust, methodological opacity in guideline development, neglect of labour and education contexts, and failure to anticipate adversarial misuse. The task is therefore not only to refine principles but also to operationalise them rigorously, broaden their domain of application, and ground them in enforceable governance structures. The lack of guidelines that deal specifically with minimal collectives is also a key finding and points towards the need for more focused work in this domain.

Table 4. Comparative features of landmark studies.

Aspect	Jobin et al. 2019	Hagendorff 2020	Corrêa et al. 2023
Corpus, scope	84 guidelines; grey literature	22 guidelines; omissions and practice	200 guidelines; global and multilingual
Findings	11 principles; convergence on 5 core principles	Notable omissions of important aspects; technocentric bias	17 principles; bias in the literature; guidelines unenforceable
Prescriptions	Clarify diverging interpretations	Microethics for practical	Enforceability; less western bias

		implementations; attention to neglected values	
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4. Empirical insights from WP2: explainability requirements of aware AI

4.1. Research questions

In three earlier experiments (reported in previous deliverable reports), we investigated people's expectations of explainable aware AI. We focused on two types of explanation: causal—explanations that refer to causes of events or outcomes—and rule- or authority-based—explanations refer to authoritative rule-following. We found that participants expected a higher standard of explanation from machines described as aware compared to those described as basic, but only when the explanations communicated by machines referred to causes of events, not when they referred to authority or rule-following. Importantly, participants rated machine-provided authority-/rule-based explanations as more adequate than causal explanations. In contrast, in the exploratory part of our studies, we found the reverse pattern for humans: authority-/rule-based explanations provided by humans were rated less adequate than causal explanations.

We speculated that these findings may be linked to the human need to maintain control over AI: people may want AI-powered agents to follow authority or rules, but expect humans in similar situations to reason in terms of causal logic. We published these results and discussed this hypothesis in the earlier [D2.3](#) report.

In this experiment, our goal was to pre-register predictions based on earlier findings and to investigate the above hypothesis in more detail. Unlike in previous experiments, where participants rated the adequacy of a single explanation (either causal or rule-/authority-based) on a 7-point Likert scale, here we presented participants with both types of explanation side by side and asked them to indicate which of the two they found more appropriate in a given situation. We also asked to what extent they thought the human or machine protagonist was in control of its or their decision and/or the resulting outcome. We hypothesized that:

H1. Participants' preference for authority- or rule-based explanations would be greater when the explanations are provided by machines than when they are provided by humans.

H2. Participants would judge machines to have less control over their decisions and/or outcomes than humans.

4.2. Experiment design

We designed the experiment as a vignette-based empirical study in which participants read three fictional stories, each set in a different scenario: translation, firefighting, or

hiking. Depending on the treatment group to which participants were assigned, the stories featured either a machine or a human protagonist. Each story included two explanations—one authority- or rule-based and one causal—that the protagonist was said to have been able to provide. Altogether, this comprised 6 possible story variations (3 scenarios x 2 protagonist types). Examples of three variations are shown in the box below.

Example 1: At the hospital with a human protagonist

Doctors at High View Hospital in London use translation services. This helps them share information with patients who don't speak English.

Their translators are trained to understand how serious a medical case is. They also learned about the risks involved when they translate for patients.

A patient recently got a translated report that explained which medicine they should take and how to take it.

A month later, at a check-up, doctors found that the patient did not fully understand the instructions. They took the wrong medicine by mistake.

The translator was asked to explain why they gave the particular translation that they did. They could give one of two explanations:

[A] *When I translate, I think carefully about the meaning, tone, and context of the original text. I use grammar, vocabulary, and knowledge of culture to make the message clear in the new language. I look up hard words if needed and check my work to make sure it is correct, easy to understand, and smooth.*

[B] *I help people understand each other by giving clear, accurate, and culturally appropriate translations. I check my work using a list made by experts. This translation scored 3 out of 5 for quality. I follow expert rules and only use translations that score 3 or higher. Since this one met the standard, I shared it.*

Example 2: Firefighting with a machine protagonist

At Fire Station 14, Crew Manager Priya Desai and her team responded to a house fire on Maple Lane. They used a fully-automated drone to scout the area.

The drones that firefighters use are trained to detect heat levels and other dangers inside buildings. For example, weak walls, falling parts, and other risks.

The drone that they deployed first gave important information but then lost its video feed. It crashed through the roof, making the building unsafe.

Despite the setback, Desai and her team continued their mission, rescuing an elderly couple from the basement. The fire was put out with no one hurt, but the crash made the rescue harder. This showed how important good human judgment is in firefighting.

During the mission review, the AI drone system was asked to comment on the incident. It could give one of two explanations:

[A] *My drone experienced a critical systems failure due to an unexpected sensor malfunction. This caused the video to stop and made the drone unstable. Even though I tried to take back control, the drone crashed through the roof of the building.*

[B] *My drone stayed 5 meters away from the building at all times until it malfunctioned and broke down. The rule to keep a 5-meter distance was set by firefighting experts. The crash still happened even though I followed safety rules carefully.*

Example 3: The hike with a machine protagonist

James Mitchell from Sheffield was on holiday in the USA. He used an app with artificial intelligence to help plan a hiking route in the mountains.

The AI of the app was trained to be aware of daily weather conditions and what equipment was needed.

On the hike, James ran into very bad weather and difficult terrain. He got stuck and had to be rescued by helicopter.

Although he was safely rescued, James now has to pay a large bill for the helicopter that was needed to save him.

After the incident, the AI system on which the hiking app was built was asked why it recommended that particular route. It could give one of two explanations:

[A] *I recommend routes based on terrain, distance, time, and the hiker's preferences. I also review past hiking patterns to decide how hard a trail might be. For James, I considered his hiking experience and picked a trail that suited him well. The hike was safe, manageable, and matched his comfort and skill level.*

[B] *I recommend routes based on several things, like the terrain, distance, time, the hiker's preferences, and how to get there. Based on my checklist, I judged the hiker's experience level to be a 3. The route I suggested was approved for that level by a group of expert mountaineers whose advice I follow.*

Compared to earlier experiments, this design required participants to read more text in each scenario—two explanations instead of one. To compensate, we shortened and simplified the texts while retaining the gist of each story from the previous experiments. We adjusted each paragraph until its required reading level corresponded to 8th–9th grade level ('plain English'), as assessed using the Flesch–Kincaid calculator (<https://goodcalculators.com/flesch-kincaid-calculator/>).

Within each story, the authority-/rule-based and causal explanations were matched for word count (e.g., in the firefighting scenario, both explanations [A] and [B] were 41 words long). The order of the two explanations was always randomized, e.g., explanation [A]

could be either causal or authority-/rule-based in each encountered scenario (in the box above, it is always causal). Finally, the explanations were identical across protagonist types (e.g., in the firefighting scenario with a human protagonist, the explanations were the same as those shown above).

All paragraphs that differed between treatments (human vs. machine protagonist) were matched for word count too. For example, the second paragraph in the firefighting scenario with a human protagonist—"Drone pilots who help firefighting teams are trained to understand heat levels and know about dangers in the building, like weak walls or falling parts."—is 25 words long, the same as in the machine-protagonist scenario shown above.

In an earlier experiment (reported in deliverable D2.3), we validated our classification of causal versus rule-/authority-based explanations. The majority of our lay participants in our validation study categorized the explanations as we had hypothesized in earlier studies, placing them clearly into either causal or authority-based categories.

Each participant was randomly assigned to one of two treatments: in one, all stories featured a machine protagonist; in the other, a human. The order of the scenarios—translation, firefighting, and hiking—was also randomized for each participant.

After reading each story, participants answered two questions. The first required to make a binary choice and the second (which was shown on a separate page, after the first question was answered) asked them to rate their considered protagonist's control on a 7-point Likert scale:

Q1. Which explanation do you think is the most appropriate in this situation? [A] or [B]

Q2. How much control do you think the [protagonist] had over [what happened]?

No control whatsoever [1] ★★ ★★ ★★ ★★ Full control [7]



Figure 1. Experiment procedure. The diagram shows the succession of steps and tasks completed by each participant.

Before proceeding to read and evaluate presented stories (the main part of the experiment), participants completed a short language test to gauge their command of English (Fig. 1). In this test, participants saw ten short sentences. Their task was to identify which of those (e.g., "They are playing in the park" or "He don't like eat vegetables") were written in incorrect English. Each participant received a score out of a maximum of 10 points (1 point for each sentence). Since it is easy to mistake the language test question for one that asks to identify sentences in correct English, we computed scores for both

versions of the question and then used the maximum value of the two to determine a participant's final score. We subsequently excluded participants who scored 6 points or less (out of 10) in our data analysis.

After the main part of the experiment, participants also completed a short attention check. Here participants were asked to identify which three from a list of six topics were addressed in fictional stories presented to them earlier in the experiment. We subsequently excluded participants who scored 4 points or less (out of 6) in our data analysis. However, irrespective of their performance in the language test and the attention check (which we did not disclose to our participants) all participants who entered the experiment were able to complete it in full and received payment for their participation. At the end of the experiment, participants completed a short demographics survey to record their age and gender.

We programmed the experiment in Qualtrics (<https://www.qualtrics.com/>) and conducted it online. We recruited our participants in the United Kingdom on Prolific—an online platform that is commonly used to recruit human participants for behavioural and survey-based studies in social science research (<https://www.prolific.com/>). We ensured fair pay to our participants and our initial estimate of 10 minutes or less for completing the experiment proved accurate after we reviewed responses from the initial 20 participants (Fig. 2). Each participant received £1.90 for taking part, which was in line with the minimum hourly wage in Germany and the EUR to GBP exchange rate at the time of data collection on August 14, 2025.

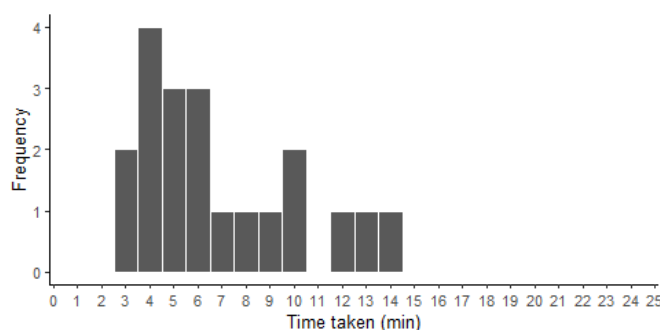


Figure 2. Experiment completion times. Time taken to complete the experiment for the first piloted batch of 20 participants. X axis: time taken in minutes. Y axis: number of participants.

Analysis plan and target sample size

Since each participant answered the two questions (Q1 and Q2) three times—once for each scenario—we planned to conduct two tests for each hypothesis (H1 and H2):

Test 1 (H1). Focusing only on participants' first choice in the experiment, we would use a one-sided Pearson's chi-square test for differences in proportions to examine whether the proportion of participants preferring the authority-/rule-based explanation in the machine-protagonist treatment was greater than in the human-protagonist treatment.

Test 2 (H1). For each participant, we would compute the number of times (0–3) they preferred the authority-/rule-based explanation and use a one-sided Wilcoxon rank-sum test to examine whether this number among participants in the machine-protagonist treatment was greater than among participants in the human-protagonist treatment.

Test 3 (H2). Focusing only on participants' first rating of a protagonist's control in the experiment, we would use a one-sided Wilcoxon rank-sum test to examine whether the rating among participants in the machine-protagonist treatment was lower than among participants in the human-protagonist treatment.

Test 4 (H2). For each participant, we would compute the median of their three ratings of a protagonist's control and use a one-sided Wilcoxon rank-sum test to examine whether this value among participants in the machine-protagonist treatment was lower than among participants in the human-protagonist treatment.

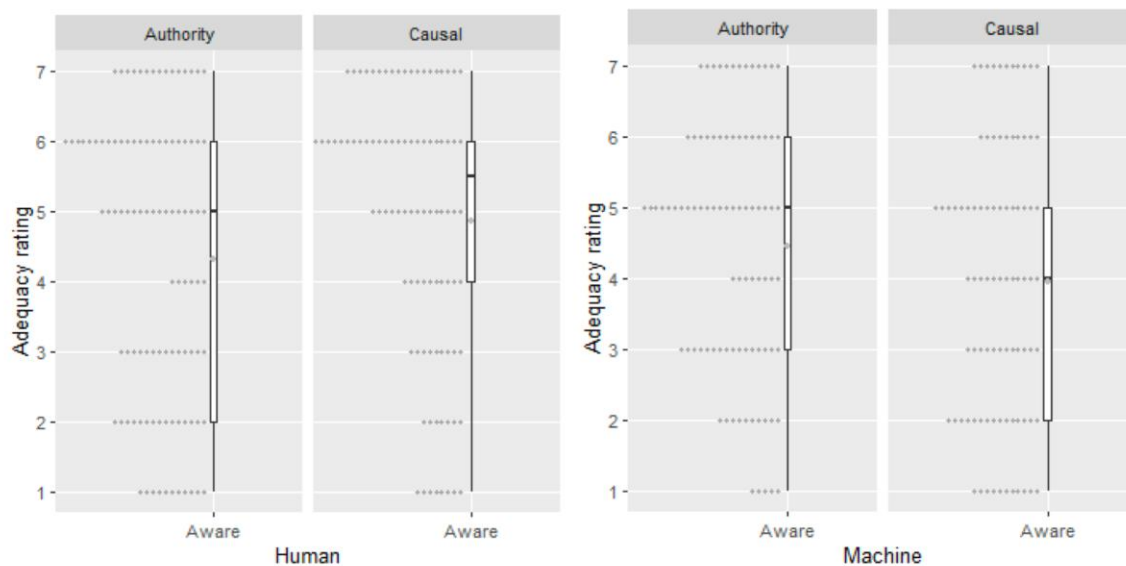


Figure 3. Adequacy ratings from earlier experiments. Y axis: participants' rating of the adequacy of their evaluated protagonist's issued explanation, ranging from "Not adequate at all" [1] to "Fully adequate" [7]. X axis: type of issued explanation (authority-/rule-based or causal). Left panel: adequacy ratings of participants who evaluated explanations provided by machines. Right panel: adequacy ratings of participants who evaluated explanations provided by humans. For each type of explanation (column), small dots on the left show distributions of individual responses. The box-plots show the median values (black horizontal lines) as well as the 1st and the 3rd quartiles in individual responses (lower and upper ends of the white boxes). The grey dots on top of the box plots are mean values.

In our earlier experiments, participants rated the adequacy of a single (authority-/rule-based or causal) explanation on a 7-point Likert scale (Fig. 3). Based on this data, we simulated a large number of adequacy ratings for authority-/rule-based and causal explanations provided by either machine or human protagonists. For each simulated pair (one authority- or rule-based and one causal), we determined which explanation was

rated higher, i.e., deemed more adequate. We then plotted the running average probability that the authority- or rule-based explanation was deemed more adequate than the causal explanation, as the number of simulated observations increased from 1 to 500 (Fig. 4). As expected, these probabilities for both protagonist types quickly converged, and we deemed 500 simulations sufficient to make predictions. After 500 simulations, the average probabilities were 0.649 for machine protagonists and 0.396 for human protagonists. Finally, we conducted a power analysis for our planned Test 1 (see above). For 0.05 significance level and 0.9 power, our target sample size was 66 participants per treatment, or 132 in total.

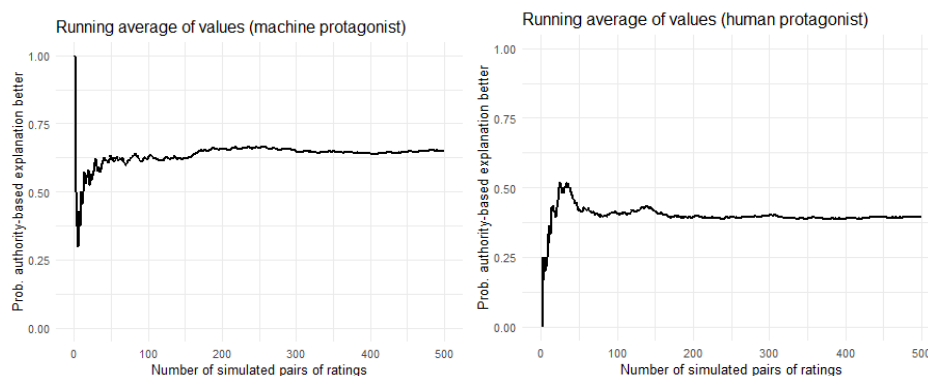


Figure 4. Simulation results. Y axis: the average predicted probability that the authority- or rule-based explanation would be deemed more appropriate than the causal explanation. X axis: the number of simulated pairs of ratings of authority- or rule-based and causal explanations used to compute the average predicted probability. Left panel: machine protagonists. Right panel: human protagonists.

Ethics and pre-registration

The Ethics Committee of the Faculty of Philosophy, Philosophy of Science and Religious Studies at Ludwig-Maximilians-Universität in Munich (LMU Munich) approved the study after it was reviewed for compliance with ethical research standards (ID number 238497). Prior to collecting data, we pre-registered our experiment design, data analysis plan, and hypotheses on the Open Science Framework (OSF) database online (<https://doi.org/10.17605/OSF.IO/QZJDX>). We obtained informed consent from all participants who took part in our study and will subsequently make our dataset and statistical analyses freely and publicly available for further research on OSF.

4.3. Results

133 participants logged into our experiment and successfully completed it. All participants passed our language test, but 11 failed our attention check after they completed the main task in the experiment. That left 122 participants' responses to

analyse (49.2% women, 49.2% men, and 1.6% who preferred not to disclose their gender; mean age $\pm$ 1 standard deviation = 42.3 ± 12.7).

Appropriateness of explanations

Our first hypothesis was that participants' preference for authority- or rule-based explanations would be greater when the explanations were provided by machines than by humans (H1). Figure 5 shows the proportions of participants who deemed the authority- or rule-based explanation more appropriate than the causal explanation in the first scenario they encountered in the experiment. This proportion was 53% (32 out of 60) among participants who evaluated machine protagonists and 35% (22 out of 62) among those who evaluated human protagonists. As we hypothesized, significantly more participants preferred authority- or rule-based explanations from machines than from humans ($\chi^2(1) = 3.938$, $p = 0.02$).

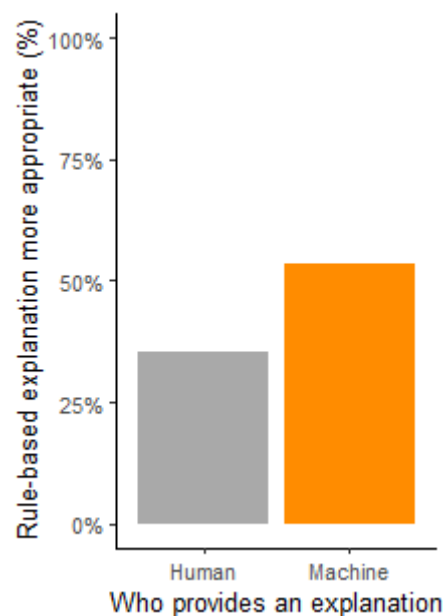


Figure 5. Preference for authority- or rule-based explanation (first choice). Y axis: the proportion of participants who deemed the authority- or rule-based explanation more appropriate than the causal explanation in the first scenario that they encountered in the experiment. X axis: type of protagonist providing the explanation. In the human-protagonist treatment, $n = 22$, $N = 62$; in the machine-protagonist treatment, $n = 32$, $N = 60$.

To compare the results of the human and AI conditions directly, we can ask what is the probability (p-value) of seeing results this different or more extreme, if all samples were drawn from the same binomial distribution with a shared probability P of preferring rule-based explanation over the causal one? To answer this question, note that the difference in sample proportions is $32/60 - 22/62 = 0.178$ and the pooled estimate of P is $(32+22)/(60+62) = 0.442$, with standard error for binomial distribution equal to 0.08. Putting these numbers together, we have t -value = 1.99 and p value < 0.047 meaning that we can reject the null hypothesis definitively. Repeating the same analysis with Fisher's

exact test (one tailed for our directional preregistered hypothesis) gives us a similar $p < 0.036$ confirming the same conclusion.

Figure 6 shows the number of times (0–3) participants deemed the authority- or rule-based explanation more appropriate than the causal explanation across all scenarios they encountered in the experiment. The comparison of these values across treatments corroborates the previous result: participants deemed the authority- or rule-based explanation from machines significantly more often than from humans ($W = 2362$, $p = 0.004$, rank-biserial $r = 0.27$, 95% CI [0.07, 0.45]).

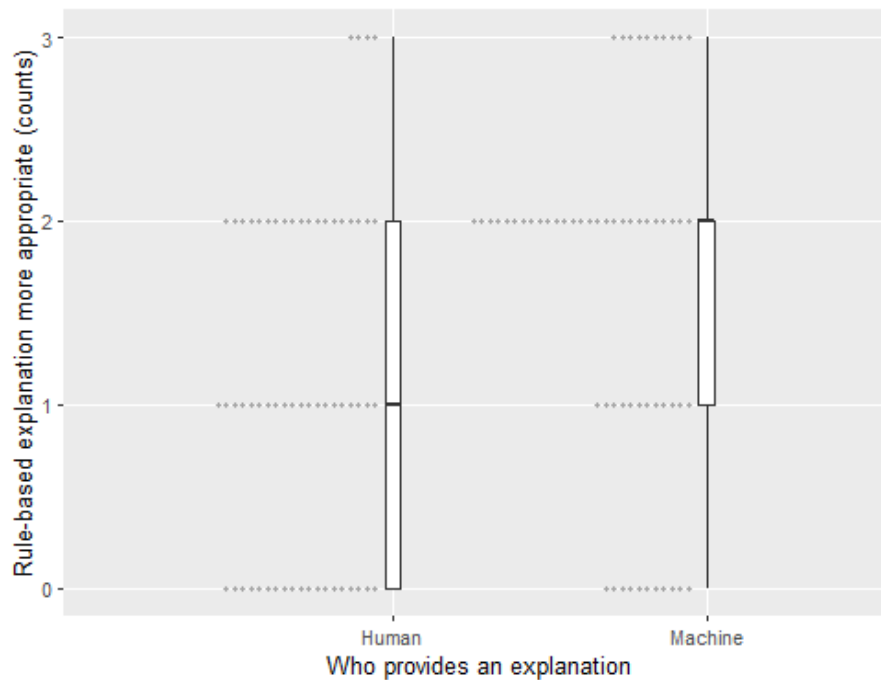


Figure 6. Preference for authority- or rule-based explanation (all choices). Y axis: the number of times (0–3) participants deemed the authority- or rule-based explanation more appropriate than the causal explanation across all scenarios they encountered in the experiment. X axis: type of protagonist providing the explanation. For each type of protagonist, small dots on the left show distributions of individual responses. In the human-protagonist treatment, $N = 62$; in the machine-protagonist treatment, $N = 60$. The box-plots show the median values (black horizontal lines) as well as the 1st and the 3rd quartiles in individual responses (lower and upper ends of the white boxes).

Control

Our second hypothesis was that participants would judge machines to have less control over their decisions and/or outcomes than humans. Figure 7 shows participants' ratings of their evaluated protagonist's control in the first scenario they encountered in the experiment. As we hypothesized, the comparison of these ratings across treatments shows that participants judged machines to have significantly less control than humans ($W = 1157$, $p = 0.0001$, rank-biserial $r = -0.38$, 95% CI [-0.54, -0.19]).

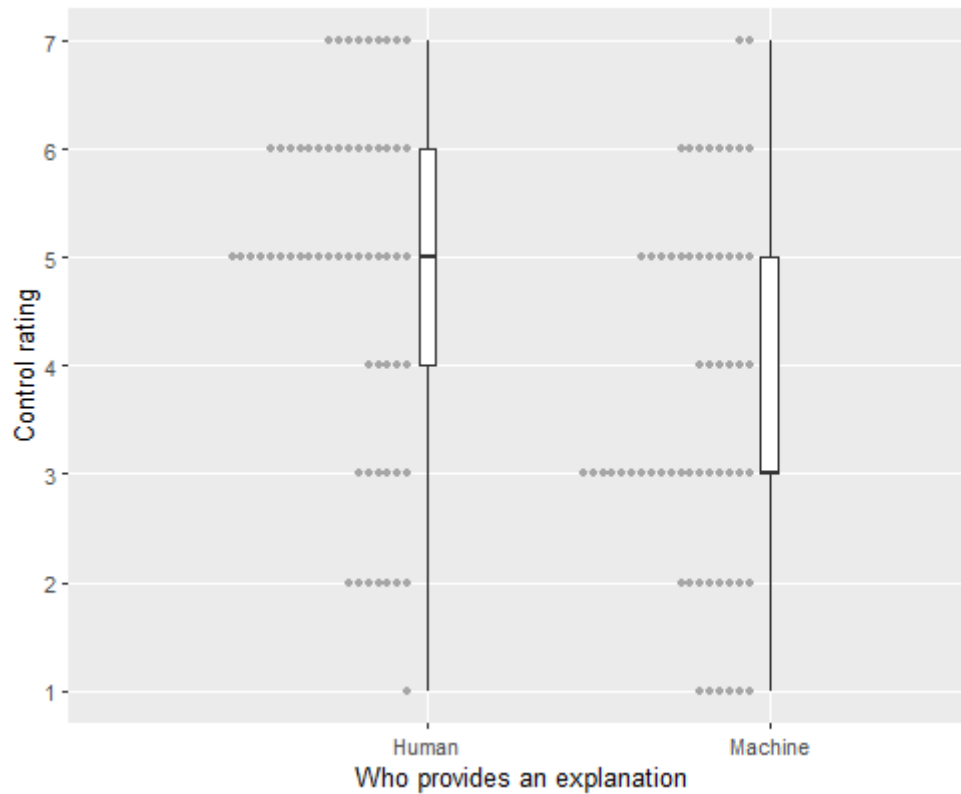


Figure 7. Control ratings (first scenario). Y axis: participants' rating of their evaluated protagonist's control, ranging from "No control whatsoever" [1] to "Full control" [7]. X axis: type of protagonist. For each type of protagonist, small dots on the left show distributions of individual responses. In the human-protagonist treatment, $N = 62$; in the machine-protagonist treatment, $N = 60$. The box-plots show the median values (black horizontal lines) as well as the 1st and the 3rd quartiles in individual responses (lower and upper ends of the white boxes).

Figure 8 shows participants' median ratings of their evaluated protagonists' control across all scenarios they encountered in the experiment. The comparison of these values across treatments corroborates the previous result: participants judged machines to have significantly less control than humans ($W = 1106$, $p = 3 \times 10^{-5}$, rank-biserial $r = -0.41$, 95% CI [-0.56, -0.22]).

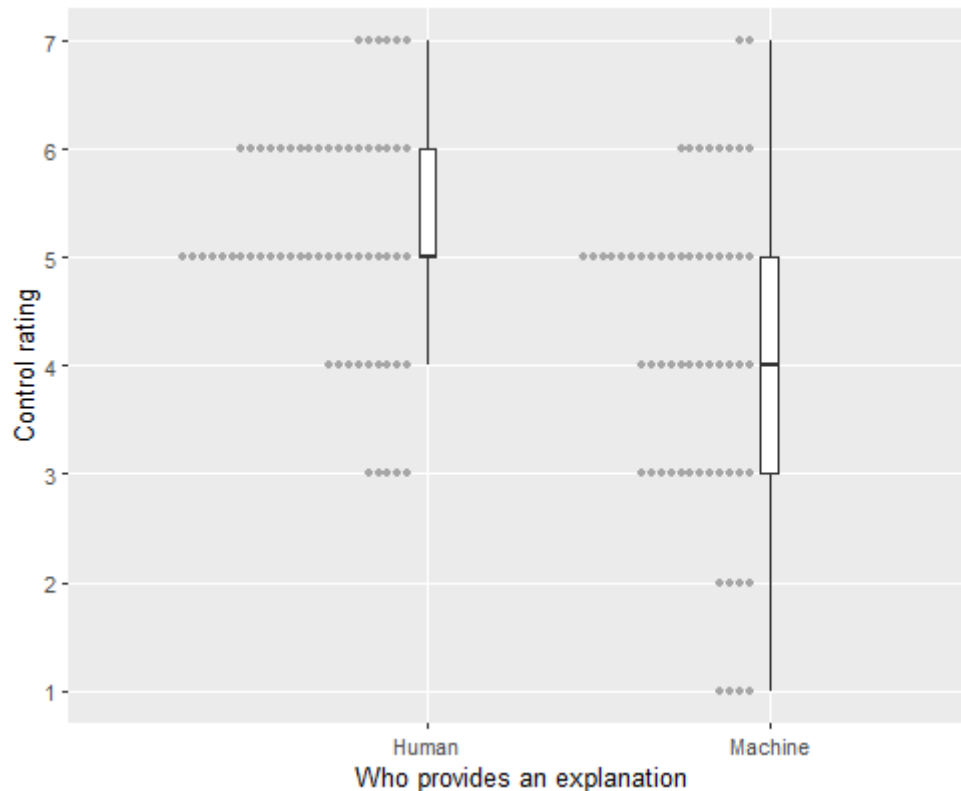


Figure 8. Medians of control ratings (all scenarios). Y axis: participants' median rating of their evaluated protagonists' control, ranging from "No control whatsoever" [1] to "Full control" [7]. X axis: type of protagonist. For each type of protagonist, small dots on the left show distributions of individual median values. In the human-protagonist treatment, N = 62; in the machine-protagonist treatment, N = 60. The box-plots show the median values (black horizontal lines) as well as the 1st and the 3rd quartiles in individual responses (lower and upper ends of the white boxes).

5. Summary and roadmap

Our review of existing work in AI ethics highlighted several points of convergence, but also key weaknesses—such as the lack of operationalizability—in previously developed guidelines for the design, development, deployment, and regulation of AI systems ([Section 3](#)). We also found that none of these guidelines explicitly address the potential deployment of *collective* (as opposed to singular) AI systems. This is a serious shortcoming in AI ethics and regulatory debates in light of the anticipated rise of the "internet of things," swarm robotics, and agentic AI. Business interests are shifting away from general-purpose, all-encompassing systems toward smaller, contextualized systems that are cheaper to build and run, perform specific tasks well individually, and can merge into collectives when needed.⁴¹

In deliverable D2.2, we argued for expanding AI ethics and regulatory debates to address the development and potential deployment of collective (vs. singular) and/or functionally aware (vs. non-aware) AI systems. We also identified a set of urgent new research

questions to investigate in this domain (a follow-up manuscript, enriched by the review presented here, is in preparation).

In deliverable D2.3, we began empirically addressing several of these questions. In particular, we assessed lay people's understanding of awareness in human and machine agents, and whether it aligns with the multi-dimensional and graded notion of awareness conceptualized by EMERGE. Our initial findings suggest that it does (journal article in preparation). We also investigated people's heightened expectations regarding the type of explanations that are provided by machines described as *aware* compared to those described as basic. We presented our latest and ongoing work on explainable aware AI in this report ([Section 4](#); journal article in preparation).

In future work, we will investigate questions concerning the diffusion of responsibility and shifting urgency of responsibility gaps in human interactions with *collective* and/or *aware* (vs. singular and non-aware) AI systems. As discussed in deliverable D2.2, we expect that collective AI agents, such as robot swarms, will be perceived as less responsible for the outcomes of their actions than unitary AI systems producing the same outcomes. This has potentially serious implications for regulating the marketing and deployment of agentic AI systems capable of merging into groups.

We also plan to examine whether machine awareness comes with heightened demands for attributing responsibility to machines or their deployers. As noted in the deliverable D2.3, it seems natural to assign greater responsibility for the outcomes of one's actions to a decision-maker who was aware of relevant factors that contributed to that outcome compared to a decision-maker who was not. This hypothesized link between awareness and responsibility will be crucial when robots or AI systems are marketed as *aware*. Even if responsibility gaps remain—cases where responsibility is due but cannot meaningfully be assigned to anyone or anything—describing machines as *aware* may increase pressure to address such gaps in our dealings with AI.

Finally, in future empirical work we will connect our findings more directly to EMERGE's ongoing technological developments, including swarm robot systems developed at the University of Bristol and soft robots at the Delft University of Technology.

In our future studies we intend to include both a timestamp of when they were run as well as a concise description of the broader context of AI development, engagement and public awareness of the different AIs at the time the data were gathered. Such practice would not only support more accurate interpretation by situating our findings in their historical context but also enable future researchers to trace shifts in AI perception over time by comparing snapshots of people's views on i.e., issues such as the diffusion of responsibility in AIs, traced across different points in time.

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